Circuit-Level Gate Noise on the VQE
Integration, Verification, and the Depth Optimum

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1 Purpose and Context

We start from a *validated* parity-constrained VQE: across the chemical-potential sweep it returns optimal parameters $\theta^*(\mu)$ that reproduce the even-parity ground state, benchmarked against exact diagonalization for energy, parity, subspace fidelity, and the edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$. The same VQE also gives a *depth diagnostic*: *noiselessly*, the subspace infidelity falls as the number of ansatz repetitions r grows, so a deeper ansatz prepares a better state.

That conclusion is only half the story on real hardware. A NISQ device builds the state with a *circuit*, and error enters *through the gates*: every entangling operation injects a fresh error channel, so the total corruption grows with the gate count and the circuit depth. This note integrates that correctly with Qiskit's noise stack and asks the quantitative question the noiseless diagnostic could only hint at: as the ansatz deepens, the prepared state improves but the accumulated gate noise worsens — so how does the edge-string diagnostic actually behave?

This note has three goals:

1. **Theory.** Treat a noisy gate as a completely positive trace-preserving (CPTP) map composed with the ideal unitary, and write the depolarizing and thermal-relaxation channels in Kraus/Pauli form.
2. **Practice.** Build a Qiskit `NoiseModel`, attach gate errors to instructions, read the edge string from a density-matrix simulation, and **verify** the integration against an independent exact gate-by-gate reference.
3. **Result.** Quantify how the edge-string diagnostic scales with ansatz depth r under per-CNOT noise — the expressibility-versus-noise trade-off.

All conventions follow the existing repository: the little-endian Jordan–Wigner qubit Hamiltonian and the edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ from `block3_core.py`.

2 Gate Errors as Quantum Channels (Theory)

2.1 The gate-then-error model

An ideal gate acts by unitary conjugation, $\mathcal{U}_g(\rho) = U_g\, \rho\, U_g^\dagger$. A noisy gate is the ideal unitary followed by an error channel $\mathcal{E}_g$,

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \qquad \mathcal{E}_g(\rho) = \sum_k K_k\, \rho\, K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I,$$

where the completeness relation guarantees the map is CPTP. This is exactly the model Qiskit Aer implements: for every instruction in the *transpiled* circuit, the associated error channel is applied immediately after that instruction. For a circuit $U = U_{g_M} \cdots U_{g_1}$,

$$\rho_{\text{out}} = \big(\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}\big) \circ \cdots \circ \big(\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1}\big)(\rho_{\text{in}}).$$

The central consequence is that **noise scales with gate count and depth**: a deeper circuit composes more error channels.

2.2 Physical origin: from Lindblad to Pauli channels

These channels are not arbitrary. A Markovian open system obeys the Lindblad equation

$$\dot{\rho} = -i[H, \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\}\Big),$$

and integrating over a finite gate duration τ produces a CPTP (Kraus) map. Isotropic, fully symmetric noise integrates to the *depolarizing* channel; energy relaxation $|1\rangle \to |0\rangle$ to *amplitude damping* (T_1); pure loss of phase coherence to *phase damping* (T_2). The practical Qiskit channels below are the gate-duration-integrated shadows of physical decoherence.

3 The Depolarizing Channel

The n-qubit depolarizing channel mixes the state toward the maximally mixed state. Qiskit's `depolarizing_error(p, n)` uses the "with probability p, replace by the maximally mixed state" convention,

$$\mathcal{E}_p^{\text{dep}}(\rho) = (1-p)\,\rho + p\,\frac{\operatorname{Tr}\rho}{2^n}\,I.$$

Using the Pauli twirl identity $\frac{1}{4^n}\sum_P P\rho P^\dagger = \frac{\operatorname{Tr}\rho}{2^n} I$ (sum over all 4^n Pauli strings), this is equivalently a random-Pauli channel,

$$\mathcal{E}_p^{\text{dep}}(\rho) = \Big(1 - p\,\tfrac{4^n-1}{4^n}\Big)\rho + \frac{p}{4^n}\sum_{P \neq I} P\rho P^\dagger.$$

We attach a two-qubit depolarizing error of strength p_{cx} after every `cx`, the dominant error on superconducting hardware.

4 Thermal Relaxation as a Gate Error (T_1/T_2)

Depolarizing noise is symmetric and structureless — the right first check, but not the real physics. The realistic upgrade attaches a *thermal relaxation* channel that depends on the gate duration τ and the device coherence times. Qiskit provides `thermal_relaxation_error(t1, t2, time)`, combining:

- **Amplitude damping** (T_1): excitation-loss probability $p_{T_1} = 1 - e^{-\tau/T_1}$, mapping $|1\rangle \to |0\rangle$;
- **Phase damping** (T_2): loss of off-diagonal coherence at rate set by T_2, with physical consistency $T_2 \leq 2T_1$ (Qiskit enforces this).

Why this matters for the Majorana diagnostic. Two effects are qualitatively distinct from symmetric depolarizing contrast loss:

- T_1 **is parity poisoning.** In the Jordan–Wigner encoding qubit occupation tracks fermion occupation, so $|1\rangle \to |0\rangle$ flips the fermion parity and pushes the state out of the even sector the diagnostic lives in.
- T_2 **kills non-local coherence.** The edge string is an endpoint-to-endpoint correlator; pure dephasing suppresses exactly the off-diagonal coherence that carries it.

Because relaxation is gate-duration-dependent and two-qubit gates are the longest, putting the dominant noise on the `cx` operations is justified from first principles.

5 Building and Verifying a NoiseModel (Practice)

5.1 Construct, attach, read out

```
from qiskit import transpile
from qiskit.quantum_info import SparsePauliOp
from qiskit_aer import AerSimulator
from qiskit_aer.noise import (NoiseModel, depolarizing_error,
                              thermal_relaxation_error)

nm = NoiseModel()
nm.add_all_qubit_quantum_error(depolarizing_error(p_cx, 2), ['cx'])

sim = AerSimulator(method='density_matrix', noise_model=nm)
qc = ansatz.assign_parameters(theta).copy()     # the validated VQE state prep
qc.save_density_matrix()
rho = sim.run(transpile(qc, sim)).result().data()['density_matrix']

O_edge = SparsePauliOp(['XZZX'], [1.0])          # L = 4 edge string
s = float(rho.expectation_value(O_edge).real)
```

`add_all_qubit_quantum_error` applies the channel after *every* occurrence of the named gate.

The transpilation gotcha. Errors attach to gate *names*, and a noise model only fires on names that survive transpilation into the simulator basis. We transpile everything to a fixed basis `['rz','sx','cx']` (`block4.py`, `BASIS_GATES`) so the `cx` the error is attached to is exactly the `cx` the circuit runs. If the target basis used `ecr` or `cz`, an error attached to `cx` would never trigger.

5.2 Precise verification: an independent exact reference

To prove the gate noise is integrated correctly — not just plausibly — we reproduce the noisy circuit a second, independent way and compare. The function `exact_reference_edge` in `block4.py` transpiles the same circuit, then evolves the density matrix *by hand*, gate by gate: it applies each gate's unitary as a `quantum_info.Operator` and, after every `cx`, applies the two-qubit depolarizing channel as a `SuperOp`. This uses none of the Aer simulator's noise machinery, so agreement is a genuine cross-check.

```
from qiskit.quantum_info import DensityMatrix, Operator, SuperOp

dep = SuperOp(depolarizing_error(p_cx, 2))
dm = DensityMatrix.from_label('0' * L)
for inst in transpile(qc, basis_gates=['rz','sx','cx'], optimization_level=0).data:
    qargs = [qc.find_bit(q).index for q in inst.qubits]
    dm = dm.evolve(Operator(inst.operation), qargs=qargs)
    if inst.operation.name == 'cx':
        dm = dm.evolve(dep, qargs=qargs)
```

Run across the whole μ sweep, the Aer `NoiseModel` edge string and this exact reference agree to $\max|\cdot| \sim 10^{-15}$ at every μ (Fig. 3, left). The gate error is therefore correctly integrated over the entire diagnostic, not merely at one density matrix.

6 Noise Integrated Through the Gates, by reps

This is the central result. The linear-entanglement `EfficientSU2` ansatz at depth r contains

$$\#\,\mathrm{CNOTs} = r\,(L-1),$$

and each carries a depolarizing channel. The measured edge string is therefore a product of two opposing factors,

$$|\langle O_{\mathrm{edge}}\rangle|_{\mathrm{meas}}(r) \;\approx\; \underbrace{|\langle O_{\mathrm{edge}}\rangle|_{\mathrm{ideal}}(r)}_{\text{expressibility (threshold)}} \;\times\; \underbrace{(1-p_{\mathrm{cx}})^{\,r(L-1)}}_{\text{gate noise, falls with } r}\;. \tag{1}$$

The two factors behave very differently, and the data make the mechanism precise. The expressibility factor is not a smooth rise but a *threshold*: at the $\mu = 0$ sweet spot the noiseless $|\langle O_{\mathrm{edge}}\rangle|_{\mathrm{ideal}}$ jumps from ≈ 0.004 at $r = 1$ (the linear ansatz simply cannot represent the $L = 4$ ground state) to ≈ 1 at $r = 2$, and stays saturated for all $r \geq 2$. The noise factor, by contrast, falls exponentially in the CNOT count $r(L-1)$ for *every* r. Their product is therefore non-monotonic — it has an **optimal ansatz depth** — but the optimum is not an interior balance point: it is the *shallowest adequately-expressive depth*, $r^* = 2$ here, the first depth that can prepare the state. Below it the diagnostic is dead; above it, every added layer is pure accumulated noise.

Plot 2 of `block4.py` shows this across the phase diagram (Fig. 1). For each depth r it prepares the VQE state across μ at fixed per-`cx` strength p_{cx}, so the only thing changing between curves is the CNOT count $r(L-1)$. The $r = 1$ curve is flat at zero (under-expressive); from $r = 2$ on, the measured sweep is pushed steadily down and the topological/trivial contrast flattens, so the same circuit depth that buys a better noiseless state costs measurable signal once the gates are noisy.

Plot 6 of `block4.py` isolates the optimum at the sweet spot (Fig. 2): the noiseless $|\langle O\rangle|_{\mathrm{ideal}}(r)$ (the expressibility step), the pure-noise envelope $(1-p_{\mathrm{cx}})^{r(L-1)}$, and the measured product on one axis. The measured curve collapses with the step at $r = 1$, peaks at $r^* = 2$, and then tracks the noise envelope down — a direct, single-μ picture of Eq. (1).

7 Verification and the Thermal Model

Plot 3 of `block4.py` combines the precise verification with the realistic noise upgrade (Fig. 3). The left panel is the exact-versus-Aer cross-check across μ described above, agreeing to

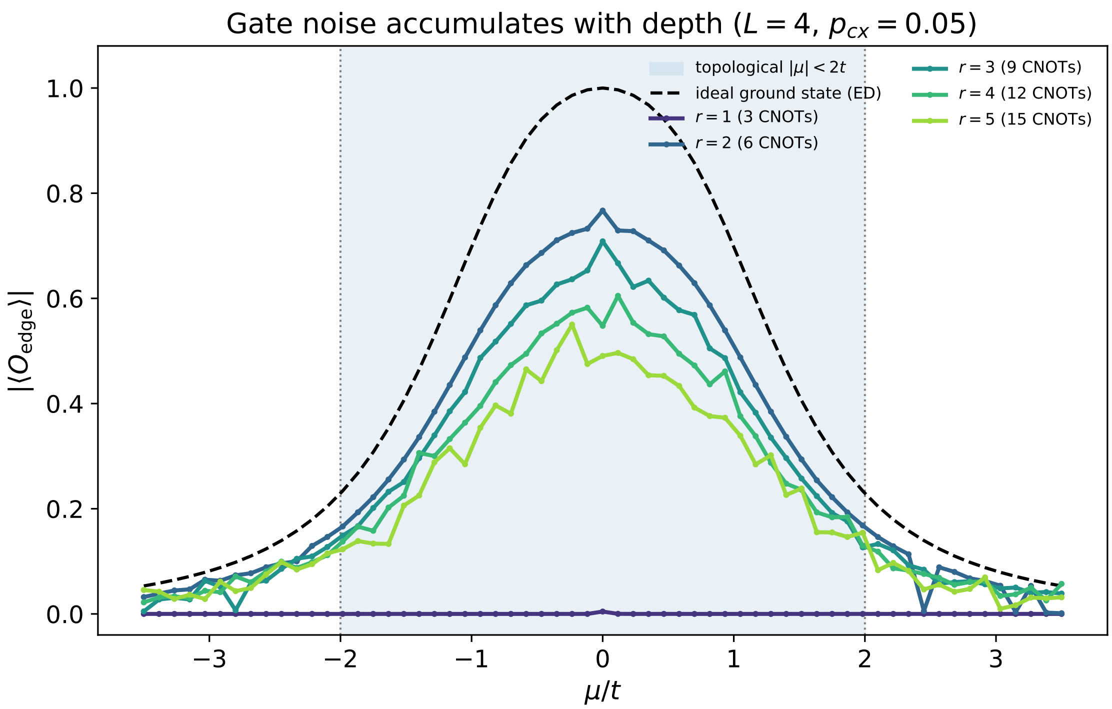

Figure 1: Edge-string sweep at increasing ansatz depth r under a fixed per-`cx` depolarizing strength. The noiseless curve (dashed) marks the topological plateau $|\mu| < 2t$; $r = 1$ is under-expressive (flat at zero), and from $r = 2$ the $r(L-1)$ noisy CNOTs push the measured diagnostic down and flatten the phase contrast.

$\sim 10^{-15}$. The right panel keeps the same VQE states but swaps the depolarizing channel for `thermal_relaxation_error`(T_1, T_2, τ) on each `cx`, with device-realistic coherence times $T_1 = 80\,\mu$s, $T_2 = 60\,\mu$s and a $\tau = 300$ ns gate. At these parameters the per-gate relaxation error is only $1 - e^{-\tau/T_1} \approx 0.4\%$, far below the deliberately large $p_{\mathrm{cx}} = 0.10$ depolarizing strength, so the thermal curve sits much closer to the ideal: realistic relaxation is the *subdominant* channel on a shallow circuit, consistent with depolarizing/readout being the first bottlenecks. The thermal channel nonetheless differs in *kind* rather than only magnitude — T_1 leaks the state out of the even-parity sector (parity poisoning) and T_2 attacks exactly the endpoint coherence the non-local string depends on — which is why it is the physically correct model to carry forward even when its present-parameter effect is mild.

8 Coherent Control Errors: $\theta^* \to \theta^* + \delta\theta$

Every channel above is *incoherent*: the ideal unitary $U(\theta^*)$ is applied exactly and a stochastic CPTP map then mixes the state, lowering its purity $\mathrm{Tr}\,\rho^2 < 1$. A real device also makes a second, qualitatively different error — the rotation angles themselves are miscalibrated, $\theta^* \to \theta^* + \delta\theta$, so the gates are slightly *wrong* rather than noisy. This is a *coherent* error: the prepared state stays *pure*, it is just rotated away from the target. The two add independently,

$$\rho_{\mathrm{out}} = \mathcal{E}_{\mathrm{gate}}\big(\, U(\theta^* + \delta\theta)\, \rho_{\mathrm{in}}\, U^\dagger(\theta^* + \delta\theta)\big),$$

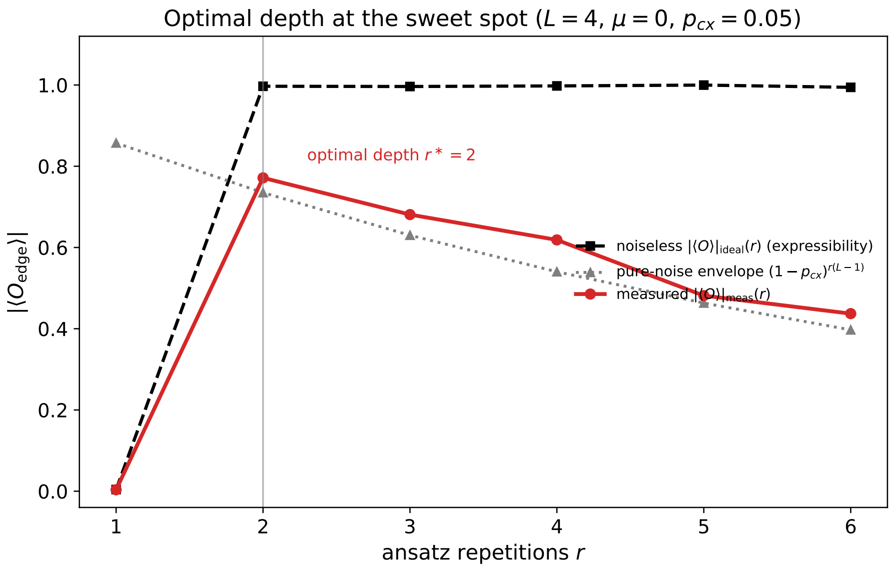

Figure 2: Depth optimum at the $\mu = 0$ sweet spot. The noiseless edge string (dashed) is a threshold — zero at $r = 1$, saturated for $r \geq 2$; the pure-noise envelope (dotted) decays smoothly; their product, the measured signal (solid), peaks at the shallowest adequately-expressive depth $r^* = 2$.

and they scale differently with depth: incoherent error accumulates roughly linearly in the gate count, while coherent miscalibration can add *or partially cancel* across layers and typically grows quadratically in the small-angle regime before it averages out. The function `perturb_theta` in `block4.py` implements both a Gaussian miscalibration $\delta\theta \sim \mathcal{N}(0, \sigma_\theta^2)$ and a systematic over-rotation (a fixed bias), and Plot 7 contrasts the two error types on the same prepared state (Fig. 4).

The distinction matters for the diagnostic: a coherent miscalibration that lowers $|\langle O_{\mathrm{edge}}\rangle|$ leaves a pure state that could in principle be corrected by recalibration or absorbed back into the VQE optimization, whereas the incoherent purity loss is genuine decoherence. The noisy-VQE optimization below is precisely the setting where the optimizer can compensate the coherent part while remaining at the mercy of the incoherent part.

9 Connection to the Majorana Edge-String Diagnostic

The edge string is the most exposed observable in the whole program because it is non-local: its value is a coherent correlation built across the chain by a ladder of two-qubit gates. In the circuit-level model every noisy `cx` injects depolarization that pulls the endpoint-to-endpoint coherence toward zero, suppressing $|\langle O_{\mathrm{edge}}\rangle|$ even when local observables still look reasonable; T_1 relaxation additionally poisons the parity sector, which is qualitatively worse than contrast loss; and the transition regions near $|\mu| \approx 2t$ are the most fragile, since the ideal signal is already changing rapidly there and little noise blurs the apparent phase boundary.

10 What This Enables Next

With gate errors integrated and verified at the circuit level, the next steps are concrete:

- **Noisy VQE optimization.** Optimize with a `NoiseModel`-backed Aer `Estimator`, so every

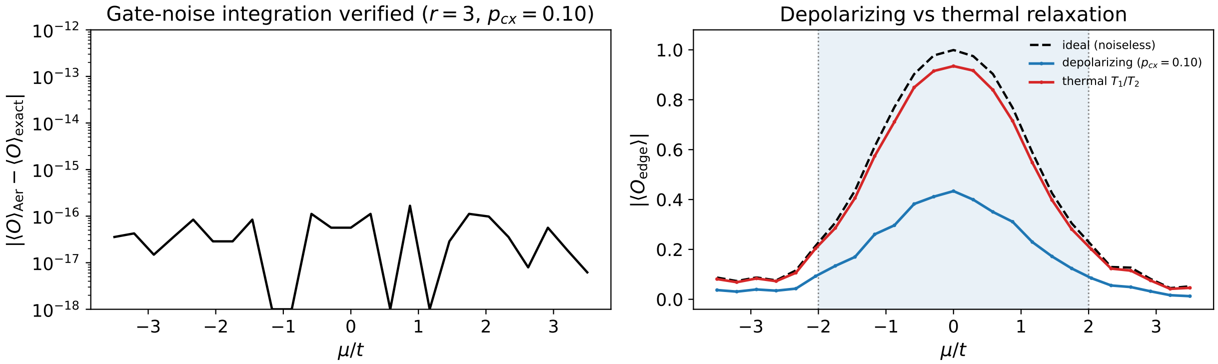

Figure 3: Left: the Aer `NoiseModel` edge string and the independent exact gate-by-gate reference agree to $\sim 10^{-15}$ across the whole μ sweep, verifying the gate-error integration. Right: depolarizing versus a physical T_1/T_2 thermal-relaxation channel at matched gate placement; the thermal model is the realistic upgrade.

cost evaluation is corrupted and the optimizer must search through noise rather than receiving a frozen validated state.

- **Backend-calibrated noise.** Use `NoiseModel.from_backend(...)` for real device T_1/T_2 and gate-error data.
- **The professor's questions.** With a faithful noisy-VQE pipeline in place, return to whether the parity is protected by topology and how the chain length interacts with the depth trade-off found here.

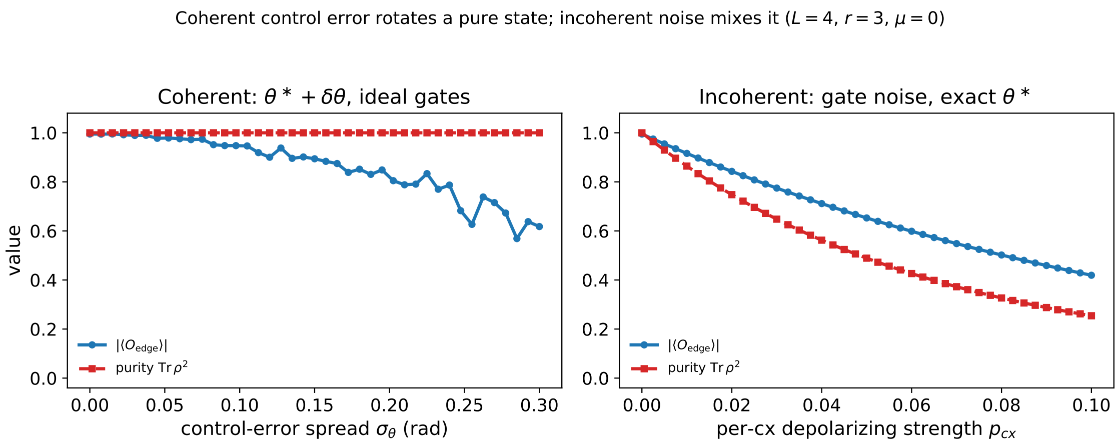

Figure 4: Left: coherent control error $\theta^* + \delta\theta$ run through ideal gates — the edge string falls as the state is rotated off target, but the purity $\mathrm{Tr}\,\rho^2$ stays pinned at 1. Right: incoherent per-`cx` depolarizing on the exact θ^* — the edge string falls *and* the purity drops. Same prepared state, $L = 4$, $r = 3$, $\mu = 0$.